Cagrilintide

0.6, 1.2, 2.4, 4.5 mg SC once weekly, dose-escalated

Cagrilintide is an experimental, long-acting amylin analogue that reduces body weight in human trials; combining it with semaglutide produces greater weight loss than either component alone. [1,5] It activates amylin receptors and the calcitonin receptor in laboratory settings. [9,10] Animal studies identify brainstem feeding circuits that help explain its likely effects on human appetite and satiety. [4,7] Its lipidated design prolongs exposure and supports once-weekly administration. [2,9]

💉 💉 💉 Human trials characterize tolerability through 68 weeks; gastrointestinal effects are common, and safety depends on the regimen and endpoint. [1,5,6]

⭐ ⭐ ⭐ Effective for weight reduction in studied populations (randomized human trials). [1,5,6]

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

Once-weekly cagrilintide has demonstrated weight loss as monotherapy and in the semaglutide combination, CagriSema. These are distinct interventions; combination outcomes should not be attributed to cagrilintide alone. [1,3,5,6]

In REDEFINE 1, adults with overweight or obesity without diabetes lost an estimated 20.4% of body weight with CagriSema, 11.5% with cagrilintide, 14.9% with semaglutide, and 3.0% with placebo at 68 weeks under the treatment-policy analysis, which includes treatment discontinuation. Both drugs targeted 2.4 mg weekly. [5]

In the sponsor's February 2026 REDEFINE 4 announcement, CagriSema 2.4/2.4 mg did not meet the primary endpoint versus tirzepatide 15 mg. Estimated weight loss at 84 weeks was 23.0% versus 25.5% under the efficacy estimand, which assumes treatment adherence. These are sponsor-reported results from an open-label comparison. [8]

Dose

Cagrilintide is administered as a once-weekly subcutaneous (SC) injection. In the phase 1b combination study, its terminal half-life was approximately 7 days with median peak concentrations reached after 24–72 hours. [2]

Doses Studied as Monotherapy

The 26-week phase 2 monotherapy trial in adults with overweight or obesity without diabetes evaluated these weekly target doses: [1]

- **0.3 mg once weekly**

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- **0.6 mg once weekly**
- **1.2 mg once weekly**
- **2.4 mg once weekly**
- **4.5 mg once weekly**

Mean weight loss increased across the tested doses, reaching 10.8% with 4.5 mg versus 3.0% with placebo under the trial-product analysis, which assumes continued treatment. The trial used dose escalation; the listed targets were not all starting doses. [1]

Dose Used with Semaglutide

The target regimen in the published REDEFINE 1 and 2 phase 3 CagriSema trials was: [5,6]

- **Cagrilintide 2.4 mg + semaglutide 2.4 mg SC once weekly**

Treatment was escalated toward this target, with flexibility for tolerability. [5,6] The published REIMAGINE 2 trial also studied CagriSema 1.0/1.0 mg weekly in type 2 diabetes. [14] In September 2026, the sponsor reported additional phase 3 results with this lower-dose combination, including a comparison with tirzepatide 5 mg; that comparison differs from the higher-dose REDEFINE 4 trial. [8,12]

The earlier phase 1b study tested cagrilintide up to 4.5 mg weekly with semaglutide 2.4 mg; 2.4 mg is therefore the principal REDEFINE target, not the highest combination dose ever studied. [2]

Dose Escalation

Gradual escalation helps manage gastrointestinal intolerance, including nausea, vomiting, diarrhea, and constipation. Tolerability varies between individuals and regimens. [1,2,6]

The clinical trial approach was: **low starting exposure → gradual escalation → individually tolerated maintenance exposure.**

In REDEFINE 1, each component began at 0.25 mg weekly and was increased at four-week intervals toward 2.4 mg by week 16; delayed escalation and dose reductions were permitted. Escalation depended on tolerability, rather than an obligation to reach the largest studied dose. [5,6]

Frequency

The approximately one-week half-life means appreciable drug remains when the next weekly dose is given. [2] The trial results support weekly administration. [1,5,6] Splitting the weekly dose into multiple administrations is a speculative community practice.

Benefits

- **Dose-dependent weight loss** — demonstrated in the phase 2 and phase 3 monotherapy trials. [1,5]

- **Likely appetite suppression and increased satiety** — supported by Cagrilintide's effects on feeding and satiety circuits in animals, alongside human weight-loss efficacy. [1,4,7]
- **Plausible earlier meal termination —** amylin reduced meal size and duration in rats, supporting this proposed contributor to cagrilintide's appetite effects. [7,15]
- **Greater weight loss with semaglutide —** established for CagriSema versus either component in randomized trials. [3,5] In REDEFINE 2, adults with type 2 diabetes lost 13.7% versus 3.4% with placebo at 68 weeks under the treatment-policy analysis. [6]
- **Plausibly complementary appetite signaling** — different amylin and GLP-1 targets provide a biological explanation for the combination benefit. [7,10]
- **Improved glycemic control with CagriSema in type 2 diabetes —** demonstrated in human trials. [3,6] In the small 32-week phase 2 trial, HbA1c fell by 2.2 percentage points with CagriSema, 0.9 with cagrilintide, and 1.8 with semaglutide. [3]

Investigated Uses

- **Obesity and/or weight-related comorbidities** — supported by human weight-management trials, alongside lifestyle intervention. [1,5,6]
- **Combination with semaglutide —** directly studied for additional weight reduction and glycemic control in the relevant trial populations. [3,5,6,14]
- **Metabolic and appetite endpoints** — animal and receptor studies investigate amylin signaling, feeding behavior, and its interaction with GLP-1 pathways. [4,7,9,10]

Cautions

- ***Pregnancy and breastfeeding —*** *safety of cagrilintide in these populations is not established. [1,5,6] Semaglutide used for weight reduction is discontinued when pregnancy is recognized and at least two months before a planned pregnancy; lactation requires an individualized assessment. [13]*
- **Active eating disorders or inadequate nutrition** — *appetite reduction makes nutritional and eating-disorder assessment a clinical precaution. [1,4]*
- **Severe gastroparesis** — *a mechanistic caution for Cagrilintide because amylin pathways can influence gastric emptying; semaglutide's prescribing information explicitly advises against its use in severe gastroparesis. [7,13]*
- ***Unstable cardiovascular disease —*** *weight-management trials do not establish safety in unstable disease or cardiovascular-event prevention. [5,6] CagriSema lowered blood pressure in a REDEFINE 1 analysis; that intermediate outcome should be distinguished from prevention of cardiovascular events. [11]*

Side effects

- **Gastrointestinal effects** — nausea, vomiting, constipation, and diarrhea occur in human trials and are commonly mild or moderate. [1,5,6]

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- **Reduced intake and dehydration** — appetite suppression or gastrointestinal symptoms can compromise food or fluid intake. [1,6,13]
- **Heart rate depends on the regimen —** in REDEFINE 1, mean pulse changed by −3.31 beats/min with cagrilintide, +0.94 with CagriSema, +1.17 with semaglutide, and −0.58 with placebo at 68 weeks. [5]
- **Hypoglycemia with other glucose-lowering medicines** — semaglutide increases risk when combined with insulin or insulin secretagogues such as sulfonylureas. This matters for CagriSema and requires medication review and glucose monitoring; it is not proof of the same risk magnitude for cagrilintide alone. [13]
- **Gallbladder and pancreatic events** — gallbladder events occurred in 2.3% with cagrilintide, 4.1% with CagriSema, and 1.0% with placebo in REDEFINE 1. Semaglutide also carries pancreatitis and gallbladder warnings when used in the combination. [5,13]

Biological mechanism

Cagrilintide activates amylin receptors AMY1, AMY2, and AMY3, which consist of the calcitonin receptor paired with receptor activity-modifying proteins RAMP1, RAMP2, and RAMP3, respectively. It also activates the calcitonin receptor without a RAMP. These findings come from receptor pharmacology experiments. [9,10]

Animal studies support central control of feeding through the dorsal vagal complex and connected circuits. The 2026 cross-species atlas studied mice, rats, and macaques and identified calcitonin-receptor/prolactin-releasing-peptide neurons also present in human brainstem tissue. Rat experiments implicated nucleus tractus solitarius Prlh signaling in sustained cagrilintide responses. These conserved circuits support a likely contribution to human satiety. [7]

Mouse receptor-deletion experiments identify AMY1 and AMY3 signaling as important contributors to cagrilintide-induced weight loss; corresponding contributions in humans are mechanistic extrapolations. [4]

Amylin and GLP-1 agonism engage partly distinct feeding pathways. This is a plausible mechanistic explanation for the greater weight loss established with Cagrilintide plus semaglutide. [3,5,7,10]

Lipidation prolongs exposure, with albumin binding proposed as a contributing mechanism. [9] The measured human terminal half-life of 159–195 hours supports once-weekly dosing but is not a direct measure of receptor occupancy. [2]

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